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1. What is the primary purpose of an Activity Diagram?

A) To model the static structure of a system

B) To model the activities of a system and the dependencies between them

C) To depict the hardware configuration

D) To design the user interface layout

E) To write pseudocode for algorithms

Answer: B) To model the activities of a system and the dependencies between them

Explanation: Slide 4 states that an Activity Diagram "models Activities of a system" and "Dependencies between activities."

2. Which of the following is a valid reason to use an Activity Diagram?

A) To choose a programming language

B) To model business processes

C) To calculate project costs

D) To hire developers

E) To name database tables

Answer: B) To model business processes

Explanation: Slide 5 lists "Modeling business processes" as a key use case for Activity Diagrams.

3. The symbol used to represent the beginning of an activity diagram is a:

A) Filled circle

B) Rounded rectangle

C) Diamond

D) Horizontal bar

E) Open circle

Answer: A) Filled circle

Explanation: Slide 8 shows the Initial/Start Node as a filled circle.

4. An action in an activity diagram is represented by a:

A) Diamond

B) Rounded rectangle

C) Filled circle

D) Horizontal bar

E) Stick figure

Answer: B) Rounded rectangle

Explanation: Slide 9 states that an action is "Represented with a rounded rectangle."

5. The text inside an action box should typically be a:

A) Noun phrase

B) Adjective phrase

C) Verb phrase in present tense

D) Past participle

E) Future tense statement

Answer: C) Verb phrase in present tense

Explanation: Slide 9 specifies that the "Text in the action box should represent an activity (verb phrase in present tense)."

6. The arrow connecting two action nodes is called a:

A) Fork

B) Join

C) Transition

D) Decision

E) Merge

Answer: C) Transition

Explanation: Slide 10 defines that action nodes are "connected by Transitions."

7. A node that represents a conditional flow of control, similar to an IF/ELSE statement, is a:

A) Initial Node

B) Final Node

C) Decision Node

D) Fork Node

E) Join Node

Answer: C) Decision Node

Explanation: Slide 13 states that a "Decision represents a conditional flow of control" and is "Similar to IF/ELSE statements."

8. On each outgoing path from a Decision Node, you must have a:

A) Final Node

B) Action Node

C) Guard Expression

D) Fork Node

E) Swimlane

Answer: C) Guard Expression

Explanation: Slide 14 and 15 emphasize that "Each branch must have a guard condition."

9. A Merge Node is used to:

A) Split the flow into parallel paths

B) Merge the flow of control from two or more branch points back together

C) Represent the end of the diagram

D) Call a sub-activity

E) Represent an iterative loop

Answer: B) Merge the flow of control from two or more branch points back together

Explanation: Slide 17 provides this exact definition for a Merge Node.

10. The symbol indicating that an action is a sub-activity defined in more detail elsewhere is a:

A) Guard condition

B) Asterisk (*)

C) Rake-style symbol

D) Thick horizontal bar

E) Diamond shape

Answer: C) Rake-style symbol

Explanation: Slide 20 shows that a sub-activity (Call Action) is "indicated by a rake-style symbol within the action symbol."

11. Forks and Joins are used to show:

A) Sequential activities

B) Activities that can occur at the same time (in parallel)

C) Only the start and end of a process

D) Conditional logic

E) The organizational structure

Answer: B) Activities that can occur at the same time (in parallel)

Explanation: Slide 21 states that "Forks and joins are used to show activities that can occur at the same time (in parallel)."

12. A key difference between a Fork and a Decision Node is:

A) A Fork has one input, a Decision has many

B) A Fork flows down all paths, a Decision flows down only one path

C) A Fork uses guard conditions, a Decision does not

D) A Fork ends the process, a Decision starts it

E) They are the same thing

Answer: B) A Fork flows down all paths, a Decision flows down only one path

Explanation: Slide 23 explicitly contrasts them: "Unlike a branch point, the control flows down all forked paths. With a branch point, the control flows down only one path."

13. A Join Node is used to:

A) Split a single flow into multiple parallel flows

B) Merge multiple parallel flows back into a single flow

C) Represent a conditional branch

D) Represent the start of an iteration

E) Call another activity

Answer: B) Merge multiple parallel flows back into a single flow

Explanation: Slide 24 defines a join as when "two or more flows of control merge into a single flow of control."

14. The rule "Every fork must have a join associated with it" ensures:

A) The diagram looks symmetrical

B) All parallel flows are synchronized before proceeding

C) Guard conditions are properly written

D) The diagram has a final node

E) Swimlanes are used correctly

Answer: B) All parallel flows are synchronized before proceeding

Explanation: This rule, stated on Slide 24, ensures that concurrent paths are properly synchronized before the main flow continues.

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15. An asterisk (*) inside an action state indicates:

A) The action is optional

B) The action is a sub-activity

C) The action may need to be performed more than once (iteration)

D) The action is performed by a system actor

E) The action is very important

Answer: C) The action may need to be performed more than once (iteration)

Explanation: Slide 29 states that "An asterisk inside an action state indicates that it may need to be performed more than once."

16. A more detailed way to show iteration, which can highlight termination conditions, is to use a:

A) Final Node

B) Decision Node

C) Fork Node

D) Note symbol

E) Swimlane

Answer: B) Decision Node

Explanation: Slide 30 recommends using a Decision Node to represent iteration as it can better show loop termination conditions.

17. Partitions in an activity diagram, often shown as swimlanes, are used to:

A) Make the diagram colorful

B) Group actions that have some common characteristic (e.g., organizational unit)

C) Represent decision points

D) Replace transition arrows

E) Indicate the end of the process

Answer: B) Group actions that have some common characteristic (e.g., organizational unit)

Explanation: Slide 34 defines an activity partition as "an activity group for actions that have some common characteristic" and notes they often correspond to "organizational units."

18. What does a thick horizontal or vertical bar represent in an activity diagram?

A) A Decision Node

B) A Merge Node

C) A Synchronization Bar (Fork or Join)

D) An Action Node

E) A Final Node

Answer: C) A Synchronization Bar (Fork or Join)

Explanation: Slide 25 states that "A synchronization bar is used to model forking and joining and is modeled as a thick horizontal or vertical bar."

19. In the "Making Coffee" example (Slide 23), "Adding Coffee" and "Steaming Milk" are connected by a fork. This means:

A) You must add coffee before you steam milk

B) You must steam milk before you add coffee

C) These two activities can happen in any order, or at the same time

D) The coffee will be rejected if the milk is steamed

E) Only one of these activities will be performed

Answer: C) These two activities can happen in any order, or at the same time

Explanation: A fork denotes concurrent flows. The order of execution for the activities on these flows is not specified and can be whatever is convenient.

20. The "Validate Membership" action leading to two paths, "[Valid Member]" and "[Else]", is an example of a:

A) Fork

B) Join

C) Decision Node

D) Merge Node

E) Initial Node

Answer: C) Decision Node

Explanation: This is a classic example of a conditional branch, represented by a Decision Node with guard conditions, as shown on Slide 18.

21. What is the purpose of a Final Node in an activity diagram?

A) To start the workflow

B) To represent a conditional branch

C) To represent the end of the workflow

D) To split the flow into parallel paths

E) To call a sub-activity

Answer: C) To represent the end of the workflow

Explanation: Slide 8 shows the Final Node and it signifies the termination of the process.

22. In the context of a fork, a "flow of control" is also known as a:

A) Swimlane

B) Thread

C) Guard

D) Action

E) Transition

Answer: B) Thread

Explanation: Slide 25 states that "A flow of control is also known as a thread."

23. Which of the following is a common mistake when drawing activity diagrams?

A) Using an initial node

B) Incorrect use of forks, branch points, merge points, and joins

C) Using verb phrases for actions

D) Placing the start node at the top

E) Using transitions

Answer: B) Incorrect use of forks, branch points, merge points, and joins

Explanation: Slide 31 lists this as a "Common Mistake," emphasizing the confusion between these control nodes.

24. When modeling a loop with a Decision Node, what is essential?

A) Using a fork node

B) Having multiple final nodes

C) Mentioning guard conditions

D) Placing it in its own swimlane

E) Making it a sub-activity

Answer: C) Mentioning guard conditions

Explanation: Slide 32, under "Common Mistakes - Loops," highlights that "Guard conditions should be mentioned" for clarity.

25. Swimlanes are typically used to represent:

A) The sequence of time

B) The cost of each activity

C) Organizational units or actors responsible for actions

D) The programming language for implementation

E) The priority of actions

Answer: C) Organizational units or actors responsible for actions

Explanation: Slide 35 shows that partitions (swimlanes) have names like "Customer" and "Order Dept," indicating who performs the actions.

26. What is the key visual difference between a Merge Node and a Join Node?

A) A Merge Node is a diamond, a Join Node is a horizontal bar

B) A Merge Node is a horizontal bar, a Join Node is a diamond

C) They look identical

D) A Merge Node is a circle, a Join Node is a rectangle

E) A Merge Node has guards, a Join Node does not

Answer: A) A Merge Node is a diamond, a Join Node is a horizontal bar

Explanation: Referring to Slide 39 (Notations), a Decision/Merge Node is a diamond, while a Fork/Join Node is a thick horizontal bar.

27. The "Take Payment" action in the DVD example (Slide 27) occurs:

A) Before the DVD is issued

B) Concurrently with updating the DVD status

C) After the flows from "Issue DVD" and "Update DVD Status" are joined

D) Only if the member is invalid

E) Instead of issuing the DVD

Answer: C) After the flows from "Issue DVD" and "Update DVD Status" are joined

Explanation: The join node synchronizes the two parallel flows ("Issue DVD" and "Update DVD Status") before the flow proceeds to "Take Payment".

28. The symbol for the initial node and the symbol for the activity final node are:

A) Both diamonds

B) Both rounded rectangles

C) A filled circle and a bullseye (circle within a circle)

D) A bullseye and a filled circle

E) Both horizontal bars

Answer: C) A filled circle and a bullseye (circle within a circle)

Explanation: Slide 8 shows the initial node as a filled circle and the final node as a bullseye.

29. In the "Buy food and drinks" example (Slide 26), the fork node indicates that:

A) You must buy food before buying drinks

B) You must buy drinks before buying food

C) Buying food and buying drinks are sequential activities

D) Buying food and buying drinks can be done concurrently

E) You can only choose to do one of them

Answer: D) Buying food and buying drinks can be done concurrently

Explanation: The fork node splits the flow into two, meaning the activities "buy food" and "buy drinks" can happen in any order or simultaneously.

30. The main success scenario in a workflow, where no errors occur, is typically modeled using:

A) Only decision nodes

B) A single sequential flow of actions and transitions

C) Only fork and join nodes

D) Only merge nodes

E) Multiple final nodes

Answer: B) A single sequential flow of actions and transitions

Explanation: The simplest form of an activity diagram, like the one on Slide 6, shows a sequential flow for the main happy path.

31. What is the purpose of a guard condition on a decision node branch?

A) To specify who performs the action

B) To define the time it takes for the action

C) To specify the condition under which that path is taken

D) To name the subsequent action node

E) To link to a sub-activity

Answer: C) To specify the condition under which that path is taken

Explanation: Guard conditions, like [valid member] and [else] on Slide 18, specify the Boolean condition that must be true for that path to be executed.

32. An activity is defined as:

A) A single action node

B) A sequence of actions

C) A final node

D) A transition

E) A swimlane

Answer: B) A sequence of actions

Explanation: Slide 9 defines an activity as "a sequence of actions."

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33. When using a decision node to model a loop, the loop continues while:

A) The guard condition on the looping path is true

B) The guard condition on the exit path is true

C) The fork node is reached

D) The join node is reached

E) The action has an asterisk

Answer: A) The guard condition on the looping path is true

Explanation: In the example on Slide 30, the loop continues [need more DVD] and exits [else].

34. What is the primary benefit of using swimlanes?

A) They make the diagram more colorful.

B) They clearly assign responsibilities to actors or organizational units.

C) They replace the need for transitions.

D) They automatically generate code.

E) They indicate the time each action takes.

Answer: B) They clearly assign responsibilities to actors or organizational units.

Explanation: Swimlanes partition the diagram based on who performs the actions, clarifying responsibilities (Slide 34, 35).

35. In the software development activity (Slide 33), the loop "Till the client is satisfied, UIE modifies the prototype" is best modeled with:

A) A fork node

B) A join node

C) A decision node and a transition back to "modify prototype"

D) A final node

E) A new swimlane

Answer: C) A decision node and a transition back to "modify prototype"

Explanation: This is an iteration. A decision node checking a condition like [client not satisfied] would loop back, while [client satisfied] would exit the loop.

36. The point where two alternative paths from a decision node come back together is a:

A) Fork Node

B) Join Node

C) Merge Node

D) Initial Node

E) Action Node

Answer: C) Merge Node

Explanation: Slide 17 defines a merge point as used to "merge the flow of control from two or more branch points back together."

37. A transition signifies:

A) The end of the process

B) The start of a parallel flow

C) The passing of control from one node to the next upon completion

D) A conditional branch

E) An organizational boundary

Answer: C) The passing of control from one node to the next upon completion

Explanation: Slide 10 states: "When the execution of the node completes, execution proceeds to the node found on the output flow."

38. The "Receive Order" action in an order processing system would most likely be placed in which swimlane?

A) Finance Department

B) Fulfillment Department

C) Customer Service Department

D) The Customer

E) The System

Answer: C) Customer Service Department

Explanation: Based on the description in Activity 6 (Slide 38), the customer service department receives the order.

39. Which node is used to ensure that multiple parallel activities都必须complete before the next activity can begin?

A) Decision Node

B) Merge Node

C) Join Node

D) Initial Node

E) Final Node

Answer: C) Join Node

Explanation: A join node synchronizes multiple incoming flows. The output flow only triggers after ALL incoming flows have reached the join.

40. The term "SwimLane" refers to the:

A) Visual notation for an activity partition

B) Another name for a transition

C) The guard condition on a decision

D) The symbol for an action

E) The initial node

Answer: A) Visual notation for an activity partition

Explanation: Slide 35 states: "Activity partition may be shown using a SwimLane notation."